The AI Data Governance Framework

A Five-Stage Control System for the Data Boundary in AI Systems

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Framework Specification, Version 1.0
14 September 2026
doi.org/10.5281/zenodo.22285047
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Framework Specification, Version 1.0

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doi.org/10.5281/zenodo.22285047

Affiliation: iSystematic Inc.
Version: 1.0
Date: 14 September 2026
Status: Public technical specification. Self-declared conformance only. Not a certification scheme.
Licence: Creative Commons Attribution 4.0 International (CC BY 4.0)
DOI (cite this): https://doi.org/10.5281/zenodo.22285047
This version: https://doi.org/10.5281/zenodo.22285048

Cite as: Khan, N. A. (2026). *The AI Data Governance Framework: A Five-Stage Control System for the Data Boundary in AI Systems. Framework Specification, Version 1.0.* Zenodo. https://doi.org/10.5281/zenodo.22285047

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Canonical name. This framework is registry entry **REG-03** of the Defensible AI Framework Registry [1], which fixes its canonical name, its relationships and its position in the architecture. It was asserted in the source book [7] as the *AI Data Governance Stack*. That former name is recorded at Appendix A here and at Appendix A.2 of the registry, and remains the correct citation for the work published under it.

Abstract

An institution asked whether a particular item of data may be inside a particular AI system usually answers from recollection. Someone remembers which extract the training set came from, someone else remembers that the extract excluded a restricted field, and a third person remembers that the exclusion was agreed with the data owner who has since left. The answer is often correct. It is never evidence.

This specification states the control system that makes the same answer come from a record. It operates in five ordered stages. **Classify** determines, per data set, its sensitivity under a declared scheme, its personal-data status, and the residency clauses that reach it. **Bound** fixes the residency rule set, the clauses the institution may not cross, each traced to a primary instrument and dated. **Prove** constructs the lineage graph from an attested source through every transformation to the point of use. **Gate** applies the quality thresholds, the minimization check and any certification claimed. **Release** admits the data with a BOE Declaration attached, or does not admit it.

The framework is the worked treatment of the **data boundary class** defined in A Pattern Language for Production LLM Platforms [2], which specifies and governs that class. Residency is named here as a member of it rather than as a class of its own. The Bound stage produces the artifact the Cross-Border AI Architecture Patterns [1] consume, and the two specifications share one schema for it, published here as `residency-rule-set-schema-v1.0.json`.

That schema is the reason this document exists now rather than later. The registry records the absence of a machine-readable classification schema as **the single largest gap** in this entry [1]. Section 8 closes it. It closes the corresponding gap in the consuming framework in the same object, because a rule set expressed once and read from both ends cannot drift. Two copies would drift.

The contribution is architectural. **No institution unconnected to the author has been observed operating the five stages as specified here**, no independent evaluation exists, and Section 11 states what would falsify the framework. It is a specification, not a certification scheme, and no conformity assessment body operates against it.

Keywords: AI governance; data governance; data residency; data lineage; data classification; evidence architecture; boundary class; machine-readable schema; regulated industries; governed production AI.

Notation

This specification uses **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** as the normative verbal forms.

- **MUST** indicates a requirement. An implementation that does not meet it is not conformant with this specification.
- **SHOULD** indicates a recommendation. Departure requires a recorded rationale.
- **MAY** indicates a permission.

Normative terms here bind an institution's engineering and governance practice. They do not state, interpret or qualify any legal obligation. Where this specification says a clause MUST be traced to a primary instrument, it is specifying the machinery. What the instrument requires is a question for the institution's counsel and its supervisor, recorded at 1.5 as a non-goal rather than as an omission.

Defined terms

Data boundary class. One of the five boundary classes defined at 2.5 of A Pattern Language for Production LLM Platforms [2], covering where data may go and what may be retained. That specification defines the class and governs it. This one is its worked treatment for institutional data governance, and **defines no new class**.

Residency clause. A dated statement, traceable to a named primary instrument, fixing where data governed by it may be held or processed. A clause is a boundary in the sense of [2]: the optimizer may not cross it. **Residency rule set** is the set of clauses in force for an estate, at a version, with a review date. It is the artifact the Bound stage produces and the Cross-Border AI Architecture Patterns consume.

Data set. The unit of classification. One data set carries one classification record. **Data domain** is the grouping within which accountability for a data set is held by one person, per 2.4.

Attested hop. One transformation between a data set and its successor, carrying a record of who attested that the transformation introduced nothing from an unattested source, and when.

Institution. The accountable legal entity. **Regulated institution** means one whose ability to operate is conditioned on its ability to reconstruct and defend a decision to an external authority.

Normative status of sections

Sections 2 through 9 are normative. Sections 1, 10, 11 and 12 are informative, except where they state a requirement using **MUST**.

---

1. Introduction

1.1 The problem

The question an authority asks about data in an AI system is narrow and hard. Not whether the institution has a data governance policy, which it does. Whether this data set, in this model, at the time this decision was taken, was data the institution was permitted to hold, permitted to move to where it processed it, and permitted to use for the purpose it used it for. Three permissions, one moment.

Three properties of AI systems make the ordinary answer fail.

Transformation destroys provenance. Data arrives, is joined, filtered, aggregated, embedded and sampled. Each step is defensible. The training set at the end has no property recording what went into it. The assurance that a restricted field was excluded is an assurance about a process. It is not an assurance about an artifact.

Residency is decided upstream of the architecture and discovered downstream of it. A clause forbidding movement is usually known to the compliance function and unknown to the team choosing where inference will run. It is violated by a deployment decision taken by people who were never shown it in a form they could act on. Nobody decided to breach it.

The person is the record. The most common evidence for a data governance decision is the memory of whoever made it. That evidence has a departure date.

doi.org/10.5281/zenodo.22285047

1.2 What this framework specifies

A control system in five ordered stages, each producing a named artifact, such that the question at 1.1 is answered from the artifacts rather than from the people. The stages are **Classify**, **Bound**, **Prove**, **Gate** and **Release**. Sections 3 through 7 specify them.

Data governance for AI is not a documentation exercise. It is the construction of artifacts that outlive the people who produced them. That is the whole claim. Everything normative in this document follows from it.

1.3 Contribution

1. **Five ordered stages**, each with a required artifact and a stated failure condition, so that a stage can be observed to have run rather than reported to have run. Sections 3 to 7.
2. **The residency rule set as a first-class artifact**, versioned, sourced, dated and expiring, rather than as a paragraph in a policy. Section 4.
3. **One machine-readable schema serving two frameworks**, with a producer profile and a consumer profile that reference each other rather than restate each other. Section 8.
4. **An explicit statement of what the schema cannot check**, at 8.5, because a validator whose limits are undocumented will be trusted past them.

1.4 Scope

Regulated institutions operating AI systems over data that is subject to at least one classification obligation, one residency clause, or one permissibility determination. The framework governs the data boundary at the **Technical Substrate** altitude of the MESA Framework [3]. That is the enforcement altitude. A control there is enforced, not described.

1.5 Non-goals

- **Not a statement of any residency or transfer rule.** This specification names no jurisdiction and states no legal obligation. Clauses are the institution's to source, date and re-verify, and 4.5 states why.
- **Not a data management body of knowledge.** It specifies no quality measurement method, storage architecture, catalogue tooling or transformation framework. It extends a discipline that specifies those, and an institution without that discipline should build it first.
- **Not a certification scheme.** No conformity assessment body operates against this specification, and Section 9 defines the only claim it supports.
- **Not a Sharia instrument.** The Halal data certification chain referenced at 6.4 is specified elsewhere, by the framework that owns it.
- **Not an empirical claim.** Section 11 states what has not been tested.

---

2. Conceptual Model (Normative)

2.1 The class this framework works, and does not define

The **data boundary class** is defined at 2.5 of [2] as the class that fixes where data may go and what may be retained, with residency rules, data-handling classes and retention exclusions named as its typical artifacts. **That specification defines and governs the class.** This framework is its worked treatment at institutional scale, and **MUST NOT be read as defining a class of its own**. The class belongs to [2].

Residency is therefore a **member of the data boundary class**, not a class beside it. That is what allows a residency attestation produced here and a retention exclusion produced by a platform control to compose: both are instances of one class under one law rather than two constructs that happen to concern data. One law reaches both.

2.2 The Boundary Invariant applied to data

Under the Boundary Invariant [2], a boundary is a clause the optimizer may not cross, and everything else is optimization. Applied to data, the division is sharp and is normative here. **A clause is a boundary.** It fixes where data may be held, where it may be processed, what may be retained and for how long, and which sources a

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determination has excluded. **Everything else is optimization.** Which store holds it, how it is partitioned, which model indexes it, how it is cached.

An institution MUST NOT express a clause as a weighting, a target or a preference. A clause expressed as a penalty term is not a clause. The failure it produces is silent. The system stays within the limit while the limit is cheap, and crosses it when it is not. A price is not a boundary.

2.3 The five stages are ordered

The stages MUST be performed in the order Classify, Bound, Prove, Gate, Release, per data set. The order is normative. Each stage consumes the previous stage's artifact, and a stage performed out of order substitutes an assumption for an artifact that does not yet exist. The order is the control.

The most common inversion is performing Bound after a deployment decision, so that the clause set is assembled to justify a topology already chosen. **A rule set assembled after the architecture is a rationale, not a boundary set.** The consuming framework refuses to select against one.

2.4 Accountability is per domain

Each data set MUST carry one named accountable data owner for its domain, and that role MUST be occupiable by one person. A committee, a function, a team or a board MUST NOT be recorded as the accountable owner.

Accountability is per domain rather than per institution for the reason the registry gives [1]: a single accountable person for all institutional data is a person accountable for nothing specific. The unit of accountability is the data set. Where one AI system draws on several domains, each retains its own owner and the deployment gate names one of them as accountable for the gate decision, per [4]. One gate, one accountable owner.

2.5 What the framework produces, and who reads it

Four artifacts leave this framework. Its value is entirely in whether they exist. The **classification record** and the **lineage graph** are read by gate one of the deployment model [4], by model risk pre-validation, and by incident reconstruction [1]. The **residency rule set** is read by the Cross-Border AI Architecture Patterns [1]. The **certification record** is read by the authority that requires the certification. Each is named in the evidence subsection of the stage that produces it.

---

3. Stage One: Classify (Normative)

3.1 The requirement

Every data set proposed for use by an AI system MUST carry a classification record before any other stage may run on it.

The record MUST state, at minimum: the data domain and the named accountable data owner; the sensitivity level under a classification scheme the institution declares; the personal-data status; the identifiers of every residency clause that reaches the data set; and the date of classification.

3.2 The institution declares the scheme

This specification does not fix a classification scheme. An institution operating under a scheme its supervisor recognizes MUST NOT be required to restate its data in someone else's levels, and a framework imposing one would be asking institutions to maintain two classifications and reconcile them.

What is normative is that the scheme is **declared, versioned and carried with the record**. A record citing a level without the scheme that defines it is not a classification record. The scheme travels with the level.

3.3 Undetermined is a state, not a default

Personal-data status MUST be recorded as one of: none, personal, sensitive-personal, or undetermined.

doi.org/10.5281/zenodo.22285047

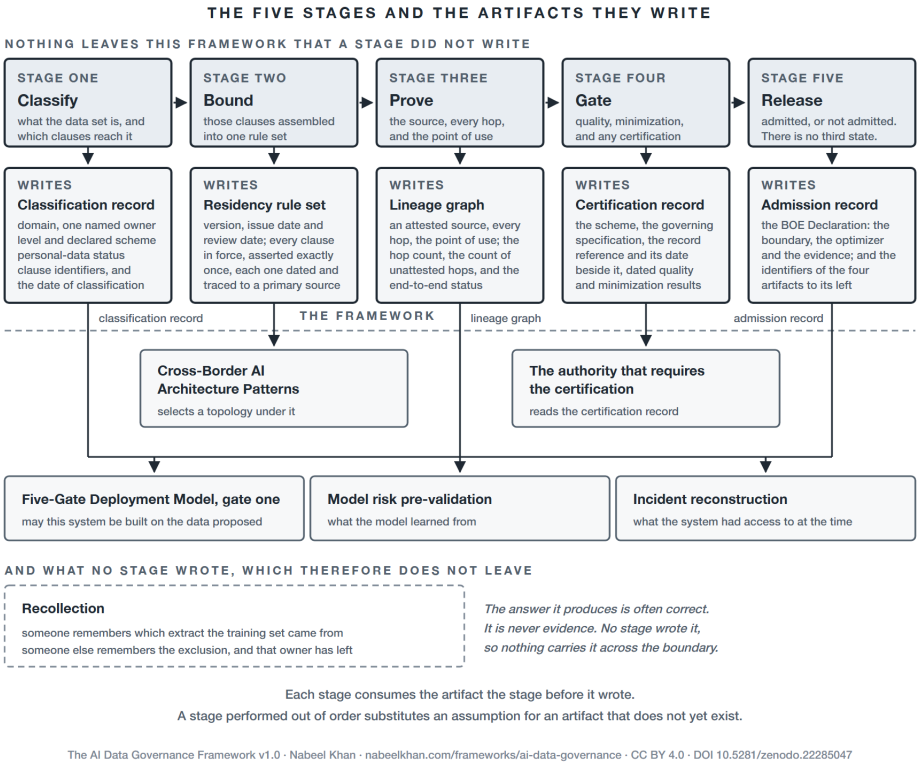


Figure 1. The five stages and their artifacts. Classify determines what a data set is and which clauses reach it; Bound assembles those clauses into the versioned, dated residency rule set; Prove constructs the attested lineage graph from source to point of use; Gate applies quality thresholds, the minimization check and any certification claimed; Release admits the data set with a BOE Declaration attached. The rule set leaves to the Cross-Border AI Architecture Patterns, which selects a topology under it; the classification record, the lineage graph and the certification record leave to gate one of the Five-Gate Deployment Model, to model risk pre-validation, and to incident reconstruction. Nothing leaves the framework that was not written by a stage. DOI https://doi.org/10.6084/m9.figshare.33420172

doi.org/10.5281/zenodo.22285047

`undetermined` is a real and reportable state, and recording it honestly is the point of having it. **A data set whose personal-data status is undetermined MUST NOT be admitted at Release.** What it MUST NOT be is the value a record carries because nobody looked, so an institution MUST record the reason a status is undetermined and the date the determination is expected.

An absent field and a field recording an unfinished determination are different claims. Only the second can be counted.

3.4 Evidence

A classification record per data set admitted to an AI system. The count of admitted data sets carrying no classification record is the framework's simplest failure measure, and an institution that cannot produce that count is not operating this stage. The measure is a count of absences.

---

4. Stage Two: Bound (Normative)

4.1 The requirement

An institution MUST maintain a residency rule set for the estate in scope, at a stated version, with an issue date and a review date, containing every residency clause in force.

The rule set is the artifact this stage exists to produce. The Cross-Border AI Architecture Patterns consume it [1]. Section 8 specifies the single schema in which it is expressed.

4.2 What a clause MUST carry

Every clause MUST carry all of the following, and a clause missing any of them is not a clause for the purposes of this framework.

Field	Why it is required
The jurisdiction it belongs to	A clause without one cannot be checked against an authority's scope
The subject it governs	Stated as the data or processing it reaches, so a data set can be tested against it
Its disposition, from the closed vocabulary at 4.3	An open vocabulary is how two documents come to describe one clause in two incompatible ways
The exposure attaching to a breach	Two clauses permitting the same movement may carry very different consequences for getting the permission wrong
A primary source: instrument, provision, citation, retrieval date	A clause whose provenance cannot be checked is an assertion
The date it was verified	An undated verification is not evidence that one occurred
The date by which it MUST be re-verified	Stated at 4.5

4.3 The disposition vocabulary is closed

A clause's disposition MUST be one of four values, and a new value MUST NOT be introduced except by a change to this section.

- **`must-not-leave`.** No data governed by the clause may leave the jurisdiction.
- **`may-move-derived-only`.** Named derived artifacts may move; the underlying data may not. The permitted artifacts MUST be named rather than implied, because a derived artifact that carries the underlying data is not a derived artifact for this purpose.
- **`may-transfer-under-mechanism`.** Transfer is permitted where a named mechanism is in force. The mechanisms and the accepted destinations MUST both be recorded.
- **`unresolved`.** The disposition has not been determined.

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`unresolved` is a blocking state and MUST NOT be treated as a permission. An undetermined clause is the state an institution is in most of the time. The failure mode is quiet. Undetermined becomes permitted because a deployment is waiting on it. Waiting is not permission.

4.4 One clause, asserted once

A clause MUST be asserted exactly once in the rule set, and every other artifact that depends on it MUST reference its identifier rather than restate its content.

This is the closure rule of the registry [1] applied to an artifact rather than to a register, and the reason is the same. A clause written down twice is a clause that can be written down twice differently. Both statements will be defensible. They will diverge under editing, and the divergence will be found by whoever relied on the wrong one. One clause, one place.

The consequence for the consuming framework is at 8.3: a selection references clause identifiers and **MUST NOT introduce a clause**. An architect finding a needed clause absent has found a defect in the rule set. The defect is returned here. Supplying the clause at the point of need is not the response.

4.5 Clauses expire, and this framework states no clause

Every clause MUST carry a re-verification date, and a clause relied on after that date is not a clause. The rule set's review date MUST NOT be later than the earliest re-verification date among its clauses, because a rule set is only as current as its most stale member.

This specification names no jurisdiction and states no residency or transfer rule. That is the reason the expiry field exists. Jurisdiction-specific rules age, sometimes quickly. A specification carrying them ages with them. It goes on reading as authoritative. The registry records the same limitation against the consuming framework in the same terms [1]: every regulatory assertion MUST be re-verified against primary sources at the time of use.

A superseded clause MUST be retained and marked rather than deleted, so that a selection made under it remains reconstructable. A rule set that deletes its history makes every prior decision undefendable at the moment it is most likely to be questioned. A deleted history cannot be produced.

4.6 Evidence

A residency attestation per jurisdiction in which data is held or processed, and the rule set itself, versioned and dated. Undated, it evidences nothing.

---

5. Stage Three: Prove (Normative)

5.1 The requirement

Every data set admitted to an AI system MUST carry a lineage graph reaching from an attested source, through every transformation, to the point of use.

5.2 The unattested hop rule

A lineage graph with an unattested hop is not a lineage graph.

The purpose of the artifact is end-to-end reconstruction. A single gap defeats it. A graph complete except at one join is not ninety percent of an answer. It is an answer that stops at the join, and the join is where a provenance question most often terminates. One gap ends the graph.

Each hop MUST record the transformation, the person who attested that it introduced nothing from an unattested source, and the date of that attestation. An automated pipeline MAY produce the record, and the attestation is still a person's, because the claim is about what the pipeline was configured to do rather than about what it did on one run.

doi.org/10.5281/zenodo.22285047

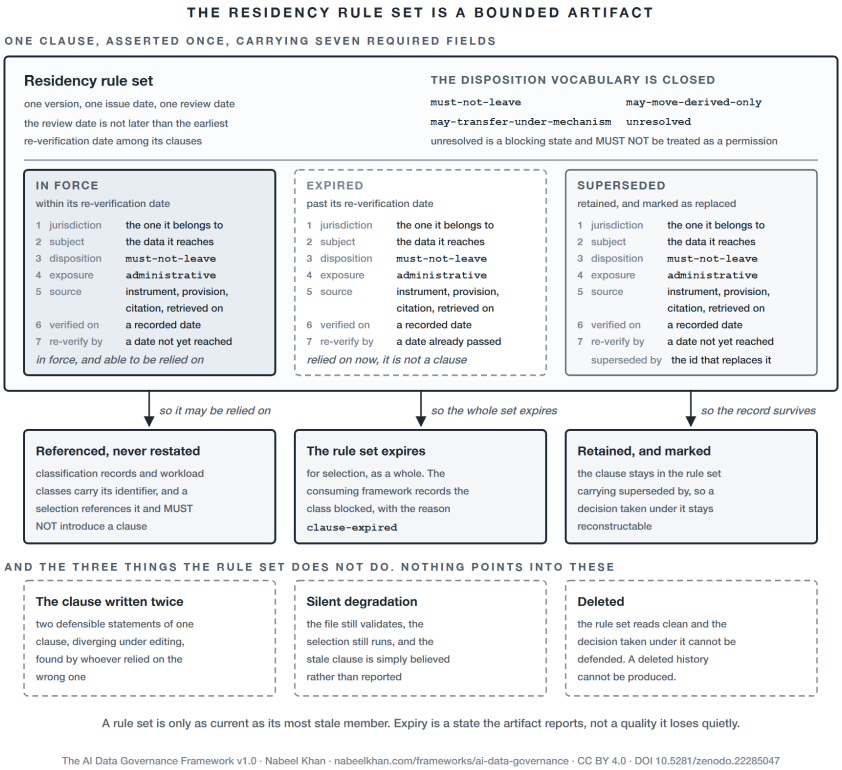


Figure 2. The residency rule set as a bounded artifact. Each clause carries a jurisdiction, a subject, one of four dispositions, an exposure, a primary source with a retrieval date, a verification date and a re-verification date. Clauses are asserted once; classification records and workload classes reference them by identifier. A clause past its re-verification date expires the rule set for selection purposes. It does not degrade it silently. A superseded clause is retained and marked so that a decision taken under it stays reconstructable. DOI https://doi.org/10.6084/m9.figshare.33420181

doi.org/10.5281/zenodo.22285047

5.3 What the graph is for

Three consumers read it. Model risk pre-validation reads it to establish what the model learned from [1]. Gate one of the deployment model reads it to establish that the system may be built on the data proposed [4]. Incident reconstruction reads it to establish what the system had access to at the time of the event [1]. Three readers, one artifact.

The third consumer is the reason the artifact must be produced continuously rather than assembled on demand. A lineage graph assembled after an incident is an account of the pipeline as it stands. Someone may have changed it after the incident.

5.4 Evidence

A lineage graph per admitted data set, with a recorded hop count, a recorded count of unattested hops, and a stated end-to-end attestation status. The count of unattested hops is reported, not concealed. A stated defect can be closed. An unstated one cannot.

---

6. Stage Four: Gate (Normative)

6.1 The requirement

Before a data set may be released, the institution MUST apply its quality thresholds, a minimization check, and any certification the data set claims. Each produces a dated result. A stage that produces no result has not run.

6.2 Quality thresholds are the institution's, and the result is not

This specification does not fix quality dimensions or thresholds. The dimensions belong to the institution's data management discipline. This framework extends that practice. It does not replace it.

What is normative is the **form of the result**: each dimension applied MUST be named, its threshold stated, its measured value stated, and its outcome recorded as pass or fail. A threshold recorded without a measurement is an intention. A measurement recorded without a threshold cannot fail.

6.3 The minimization check

The institution MUST record, per data set, whether the data admitted is limited to what the stated purpose requires, with a date and an outcome of pass, fail or conditional. Where the outcome is conditional, the conditions MUST be stated and MUST be testable.

The check sits at this stage rather than at Classify deliberately. Minimization is a judgment about a purpose. The purpose is not known at classification time. Classify is too early.

6.4 Certifications are recorded here and specified elsewhere

Where a data set claims a certification, the Gate stage MUST record the scheme, the identifier of the specification governing that scheme where one exists, a reference to the certification record, and its date.

This framework specifies no certification scheme. It records that one was maintained and where the record is, and a scheme's own requirements are the governing specification's to state. This framework records, it does not certify.

The Halal data certification chain. Where an institution operates under a Sharia Supervisory Board holding binding approval power, the chain is threaded into this framework by the Sharia AI Compliance Framework [5] and is **fully specified at section 5 of that document**, which governs it. This specification cites it, does not restate it, and adds no requirement to it. That specification governs the chain.

Normative disclosure, carried from the specification that owns the construct [5] and from registry entry REG-03 [1]. The Halal data certification chain is an original construct of that framework. **No standard-setter recognizes such a chain as of the date of this specification.** It MUST be presented as author methodology and MUST NOT be presented as an industry or supervisory requirement. This clause MUST accompany any presentation of the chain, including one made through this framework.

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6.5 Evidence

A dated quality result per dimension applied, a dated minimization result, and a certification record per certification claimed.

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7. Stage Five: Release (Normative)

7.1 The requirement

A data set is admitted, or it is not. There is no partial admission. An institution MUST NOT operate a state in which data is used while its admission is pending.

Admission MUST NOT be recorded unless all of the following hold. The personal-data status is determined. The lineage graph is attested end to end. The minimization check has a recorded result of pass, or of conditional with every condition closed. Every clause the data set engages is present in the rule set and is within its re-verification date. And a BOE Declaration is attached.

7.2 The BOE Declaration

Every admission MUST carry a BOE Declaration in the shape defined at 2.10 of [2]: the **B**oundary the admission fixes, the **O**ptimizer it frees, and the **E**vidence proving the boundary held.

For a data admission the three are short. The boundary is the clause set the data set engages. The optimizer is everything about the estate the clauses do not fix. The evidence is the classification record, the residency attestation and the lineage graph.

The declaration is what makes this framework's output joinable with the rest of the governance family. A gate record, a validation record, a residency attestation and a screening attestation are all instances of one three-part declaration, which is why an incident reconstruction consumes all four without a translation layer [1]. **Two controls that emit evidence in the same shape can be composed. Two that do not are merely adjacent** [2].

7.3 Evidence

An admission record per admitted data set, carrying its BOE Declaration and the identifiers of the artifacts the four previous stages produced. One record carries the other four.

---

8. The Machine-Readable Layer (Normative)

8.1 Why one schema serves two frameworks

The registry records two `missing` artifacts that are, on inspection, one artifact seen from two sides. This framework lacks a machine-readable classification schema, recorded as the single largest gap in its entry [1]. The Cross-Border AI Architecture Patterns lack a decision instrument, so that the method cannot be applied consistently by two architects working independently [1].

The rule set is the only thing that passes between them [1]. A schema published twice, once by the producer and once by the consumer, would be two statements of one artifact, which is the defect the registry's closure rule exists to prevent. **The schema is therefore designed once and published once.** One file carries a producer profile and a consumer profile. One artifact, one schema.

8.2 The file

doi.org/10.5281/zenodo.22285047

File	What it is
`residency-rule-set-schema-v1.0.json`	The JSON Schema, draft 2020-12, for a residency rule set and for the selection made against it. Published with this specification and cited by the consuming specification, which does not republish it

The file is part of this deposit and is licensed CC BY 4.0 with the rest of it. It may be served, embedded and validated against without further permission.

8.3 Two profiles, one object

The `rule_set` object is authored by this framework at the Bound stage. It carries the declared classification scheme, the jurisdictions and their authorities' examination scopes, the clauses, the classification records, the workload classes and a BOE Declaration.

The `selection` object is authored by the Cross-Border AI Architecture Patterns, is optional, and is specified by that framework. A rule set is a complete artifact with no selection attached. A selection with no rule set is not a selection.

The selection MUST reference clause identifiers and MUST NOT restate clause content. This is 4.4 expressed in the schema. It is what makes the two frameworks incapable of describing one clause in two ways. The schema enforces the closure rule.

8.4 Validation requirements

A rule set MUST satisfy all of the following, and a build producing a file that fails any of them MUST NOT be relied on.

1. It validates against `residency-rule-set-schema-v1.0.json`.
2. Every clause identifier referenced by a classification record, a workload class or a selection **resolves to a clause present in `rule_set.clauses`**.
3. Every clause identifier is unique, and every data set and workload class identifier is unique.
4. Every workload class's stated `clauses_engaged` equals the union of the clauses engaged by its data sets. **The union is stated rather than computed silently, so that a reader can check the computation.**
5. `rule_set.review_by` is not later than the earliest `re_verify_by` among the clauses.
6. Every accountable data owner names one person. See 8.5.
7. No admitted data set carries an unattested hop, an undetermined personal-data status, or a clause past its re-verification date.

8.5 What the schema cannot check, stated because it will otherwise be trusted past its limits

A schema validates shape. Of the seven requirements at 8.4, the file enforces the first and two clauses of the seventh. **The remaining five are not shape.** Four of those five are a validator's obligation, and one is a person's judgement that no file will ever carry. A sixth limit sits outside the seven altogether, and it is the one most worth doing. An institution relying on validation alone will believe it has checked all seven.

Four are arithmetic the file cannot perform. Requirement 2 asks whether an identifier resolves, requirement 3 whether identifiers are unique across a document, requirement 4 whether a stated union equals a computed one, and requirement 5 whether one date precedes the earliest of many. JSON Schema expresses none of them: the first needs cross-document resolution, and the other three need a computation over the instance rather than a constraint on its shape. A build MUST therefore run these four as validator checks alongside schema validation, and a build that runs only the schema has checked one requirement of seven and part of another. Requirement 2 is stated at length below because it is the one whose failure is silent.

It cannot tell a person from a committee. This is requirement 6, and it is the one a validator cannot rescue either. The `accountable_data_owner` object requires a name and a role, and a committee's name may be written into it. The schema accepts it. The requirement at 2.4 is enforced by review, and this document says so rather than leaving the impression that the file enforces it. A validator is not a reviewer.

doi.org/10.5281/zenodo.22285047

It cannot resolve an identifier. A selection may reference a clause identifier that does not exist, and the schema accepts it, because JSON Schema carries no cross-document reference resolution. Requirement 2 of 8.4 is a validator's obligation, not the schema's.

It cannot tell a real primary source from a plausible one. This is the limit that sits outside the seven. The `source` object requires an instrument, a provision, a citation and a retrieval date. It cannot establish that the instrument exists or that the provision says what the clause reports, and no requirement at 8.4 asks it to, because no file could carry the answer. **That is a human obligation.** It is the one most worth doing, and it is named here rather than left out on the grounds that the list did not cover it.

The three limits stated at length above were established by running the schema against instances constructed to violate each rule and against instances constructed to pass. All three passed validation while violating this specification. The four arithmetic limits are not a test result and are not offered as one: JSON Schema has no construct for cross-document resolution, for uniqueness by key across array members, for equality between a stated and a computed set, or for ordering a value against the minimum of many. They follow from the language, not from an experiment.



Figure 3. The machine-readable layer, and what it does not reach. One schema file carries two profiles. The rule set is authored by this framework at the Bound stage; the selection is authored by the Cross-Border AI Architecture Patterns and references clause identifiers rather than restating clauses. Seven validation requirements sit above the schema, and the file reaches only the first and two clauses of the seventh. The remaining five are drawn outside its reach and separated by what closes them: four are arithmetic a validator MUST perform, and one, that an accountable owner is a person rather than a body, is a judgement no file will carry. A sixth obligation is drawn outside the seven altogether, because no requirement asks the file to establish that a cited primary source exists and says what the clause reports. The figure exists so that a reader who trusted a passing validation sees how much of the list that pass did not touch. DOI https://doi.org/10.6084/m9.figshare.33420178

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9. Conformance (Normative)

9.1 Self-declared

Conformance is **self-declared**. No conformity assessment body operates against this specification and no third party is accredited to assess a claim made under it.

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9.2 What conformance requires

An institution claiming conformance for a stated scope asserts all of the following for the period claimed.

1. Every data set used by a system in scope carries a classification record citing a declared scheme, per 3.1 and 3.2.
2. A versioned rule set exists for the estate, every clause carries the fields at 4.2, and no clause relied on is past its re-verification date, per 4.5.
3. Every clause is asserted once and every dependent artifact references it by identifier, per 4.4.
4. Every admitted data set carries a lineage graph attested end to end, per 5.2.
5. Quality results, minimization results and certification records exist with dates, per 6.5.
6. Every admission carries a BOE Declaration, per 7.2, and no data is in use that has not been admitted, per 7.1.
7. Every accountable data owner named is one person, per 2.4.

9.3 What accompanies a claim

A published claim MUST carry the version claimed against, the date of the evidence, the scope of systems and domains covered, and the normative disclosure at 6.4 where a Halal data certification chain is claimed.

9.4 What it does not mean

- It does not mean the institution complies with any statute, supervisory expectation or standard. The framework organizes evidence; whether that evidence satisfies an authority is the authority's determination.
- It does not mean the data is correctly classified. It means classification records exist and can be examined.
- It does not mean the institution has been assessed. Nobody assessed it.
- It does not mean a certification exists. None does.

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10. Related Work

Management system standards. ISO/IEC 42001 [6] specifies a management system for artificial intelligence at the institutional altitude. It requires that data used by AI systems be governed and does not specify the artifacts by which the governing is evidenced. This framework specifies those artifacts. It is an extension, not an alternative.

Data management bodies of knowledge. The established disciplines specify quality dimensions, cataloguing, stewardship and lifecycle. This framework depends on them and does not restate them: 6.2 requires a threshold and a measurement without fixing either. It governs the boundary and inherits the rest.

The engineering family. A Pattern Language for Production LLM Platforms [2] defines and governs the data boundary class and the Boundary Invariant this framework works under, per the precedence rule at 2.6 of the registry [1]. The class is not this framework's to move.

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11. Limitations, What Has Not Been Tested, and Falsification

11.1 No institution unconnected to the author has been observed operating the five stages

There is no implementation profile, no scored instrument, and no published result for the framework as specified here. Everything in Sections 2 through 8 is architectural. The schema at Section 8 has been validated against constructed instances, which establishes that the file behaves as this document says it behaves. A validated file is not a populated one.

11.2 The attestation cost has not been measured

The framework's cost sits almost entirely in the per-hop attestation at 5.2, and that cost has not been measured anywhere. An institution with many hops and frequent change may find the attestation burden dominates every other part of the framework. This is stated as an unmeasured quantity rather than as a manageable one, because the second claim would require a measurement nobody has made.

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11.3 The rename is recent and the schema is new

The framework was published as the *AI Data Governance Stack* [7] and renamed by the registry [1]. The former name has prior use and the current name has little. Separately, **`residency-rule-set-schema-v1.0.json` has no adopters**, A schema with no adopters is a proposal. It proposes how two frameworks should exchange an artifact, and proves nothing about whether they can. Adoption is the missing evidence.

11.4 The schema's limits are real

Stated at 8.5 and repeated here, because a reader who trusts the validator will not have read 8.5. The schema cannot tell a person from a committee, cannot resolve an identifier, cannot check uniqueness across a document, cannot compare a stated union against a computed one, cannot order a date against the earliest of many, and cannot establish that a cited instrument exists. **A passing validation satisfies one of the seven requirements at 8.4 and part of a second. It satisfies none of the remaining five, and says nothing at all about the sixth obligation that sits outside the list.** A pass is not a check.

11.5 The disclosure at 6.4 is inherited and unresolved

The Halal data certification chain remains unrecognized by any standard-setter [5]. This framework carries the disclosure. It does not resolve the condition it discloses.

11.6 Single-author instrument, with a conflict

This specification has one author, has not been peer reviewed, and has been reviewed by no institution. The author operates a consulting practice applying it, which is a conflict of interest stated here rather than in a footnote.

11.7 What would falsify the framework

The framework is falsified if institutions maintaining attested lineage to the point of use are found unable to answer a data provenance question from an authority any faster, or any more defensibly, than institutions maintaining classification alone.

This is the falsification condition the registry records for REG-03 [1], restated without weakening. The framework's cost sits in the attestation. Evidence that the attestation adds nothing to the answer would remove its justification. The observation is available to any party with access to two institutions and one authority's question. It does not require the author's cooperation.

Two narrower findings would refute parts of it. If a rule set carrying dispositions from the closed vocabulary at 4.3 cannot express a clause an institution actually operates under, the vocabulary is wrong and Section 4 does not survive in its current form. And if institutions assembling a rule set after choosing a topology produce selections indistinguishable from those that assemble it first, the ordering requirement at 2.3 is discipline without effect. Order would then be habit.

11.8 Research questions

1. Is per-hop attestation achievable at the scale of a production training pipeline, and at what cost per hop?
2. Does an attested lineage graph measurably shorten the time an institution takes to answer a provenance question from an authority?
3. Can the four dispositions at 4.3 express the residency clauses institutions in fact operate under, across a sample of jurisdictions large enough to be interesting?
4. Does a machine-readable rule set produce more consistent topology selections between independent architects than a prose rule set does?

Institutions and researchers willing to engage with any of these are invited to make contact through the ORCID record in the front matter. Question four is the one this framework and its consuming framework most need answered. Neither can answer it alone.

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12. Versioning

The major number changes when a change could make a previously conformant institution non-conformant: a requirement in Section 9 is added or tightened, a stage's minimum evidence changes, the disposition vocabulary at 4.3 changes, or a required field is added to the schema. **The minor number** changes for additions and clarifications that cannot.

A change to the schema's required fields is a major change to this specification, because the schema is normative and existing rule sets would cease to validate. A schema version is pinned to a specification version and MUST NOT be revised independently of it.

Where this specification and registry entry REG-03 [1] describe the same relationship, **the registry governs**, per 11.3 of that document. A disagreement discovered between them SHOULD be reported so that the next version of one or the other removes it, and MUST NOT be settled here.

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13. References

Every reference to a work outside this corpus was verified against a live primary source, each with a deliberate failing control, because a check that cannot fail proves nothing. The date of verification is recorded at Appendix B.

[1] Khan, N. A. (2026). *The Defensible AI Framework Registry: Canonical Names, Definitions and Relationships for the Governed Production AI Discipline. Registry Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170112

[2] Khan, N. A. (2026). *A Pattern Language for Production LLM Platforms: Governed Routing, Agent Orchestration, and AI-Native Delivery. Pattern Language Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109864

[3] Khan, N. A. (2026). *The MESA Framework: A Four-Altitude Diagnostic Model for Institutional AI Governance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22109836

[4] Khan, N. A. (2026). *The Five-Gate Deployment Model: A Deployment Discipline for AI Systems. Deployment Model Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170122

[5] Khan, N. A. (2026). *The Sharia AI Compliance Framework: A Dual-Authority Governance Architecture for Islamic Finance. Framework Specification.* Zenodo. Concept DOI https://doi.org/10.5281/zenodo.22170143

[6] International Organization for Standardization and International Electrotechnical Commission (2023). *ISO/IEC 42001:2023, Information technology. Artificial intelligence. Management system.* First edition, December 2023. Cited as the institutional-altitude management system this framework does not duplicate.

[7] Khan, N. A. (2026). *AI Governance and Compliance Frameworks for the Middle East: The Enterprise Playbook.* First Edition. iSystematic Inc. ISBN 978-1-0678960-1-0

A note on what is not cited. No statute, regulation or supervisory instrument is cited in support of any proposition here, because this document advances no proposition about what any of them requires. Clauses are the institution's to source and date, per 4.5, and a specification supplying them would be supplying the one thing it has no standing to supply and the one thing that ages fastest.

Appendix A. Naming Crosswalk: The Seven Layers of the Source Treatment

Recorded so that work published under the former name and the former structure remains citable. **A former name is not an error in the work that used it**, and neither is a former structure.

A.1 The name. The framework was asserted as the *AI Data Governance Stack* [7] and is renamed the AI Data Governance Framework™ by the registry at 6.4 [1]. The former name implied a technology stack; the content is a control taxonomy. The former name remains valid for citation of the earlier work.

A.2 The structure. Chapter 13 of [7] organizes the subject as **seven layers** spanning the data lifecycle. This specification organizes it as **five ordered stages**. The two are not two accounts of one structure. The layers are concurrent disciplines operated in parallel; the stages are ordered operations performed per data set, and the ordering at 2.3 is normative here and absent there. **The registry governs**, per 11.3 of [1], and this specification is written to it.

The crosswalk below records where each layer's subject is treated here. It is a reading aid and is **not normative**.

Source treatment, chapter 13 of [7]	Treated in this specification at
Layer 1. Sources and Collection	Classify (Section 3) for the record; Bound (Section 4) for the clauses reaching the source
Layer 2. Classification and Cataloging	Classify (Section 3). The five-level pyramid becomes one declared scheme among many permitted ones, per 3.2
Layer 3. Curation and Preparation	Prove (Section 5) for the per-hop attestation; Gate (Section 6) for quality and minimization
Layer 4. Training Data Governance	Gate (Section 6) and Release (Section 7)
Layer 5. Inference Data Governance	Not governed here. Inference-time monitoring belongs to the deployment model at gate five [4] and to model risk [1]
Layer 6. Model Output and Feedback Data	Governed only where feedback data re-enters as a data set, at which point it passes all five stages
Layer 7. Audit, Retention, and Deletion	Retention is a member of the data boundary class [2] and is treated at 2.1 and 2.2 as a clause. Deletion verification is not specified here

A.3 Two claims of the source treatment that this specification does not carry. The source treatment fixes a five-level classification pyramid; 3.2 declines to fix any scheme. The source treatment presents the seven layers as concurrent; 2.3 makes the five stages ordered. Both are changes rather than restatements, and both are recorded here so that a reader holding the book is not left to reconcile them alone.

A.4 The Halal data certification chain. The source treatment presents the chain as four steps within chapter 13. It is now **specified at section 5 of [5]**, which governs it. This specification cites it at 6.4 and does not restate it. Where the source treatment and [5] differ, [5] governs.

Appendix B. Publication Provenance

B.1 What this document is. The first standalone specification of the AI Data Governance Framework, not peer reviewed. On deposit it receives a DOI, a timestamp and citability. **Deposit confers no standard status, and nothing in this document uses the word standard for itself.** No preprint identifier exists and no preprint relation is asserted in its metadata.

B.2 Relationship to prior publication. The framework was published as threads through chapter 13 of [7] under the former name. This document is its first statement as five ordered stages, the first statement of the residency rule set as a versioned and expiring artifact, and the first publication of a machine-readable schema for its outputs.

B.3 What this deposit closes. Registry entry REG-03 records a `missing` limitation: no machine-readable classification schema has been published, described as the single largest gap in the entry [1]. `residency-rule-set-schema-v1.0.json` is that artifact, and it closes the corresponding limitation of registry entry REG-09 in the same object, for the reason at 8.1.

B.4 Publication order. Published after the registry [1], the pattern language [2], the MESA Framework [3], the Five-Gate Deployment Model [4] and the Sharia AI Compliance Framework [5], all cited at their concept DOIs, which always resolve to the latest version. It publishes ahead of the Cross-Border AI Architecture Patterns specification, which cites the schema published here and does not republish it. That ordering is required rather than incidental: the other specification prints this record's concept DOI, and a reserved DOI does not resolve until the record it belongs to is published.

doi.org/10.5281/zenodo.22285047

B.5 Verification method and date. Every external reference in Section 13 was checked against a live primary source with a deliberate failing control, on **14 September 2026**. The single external reference is [6]; its publisher refused an automated request, so its edition and date were confirmed from the IEC webstore, IEC being one of the two issuing bodies named in the reference, and a deliberately wrong edition string was confirmed absent. Every corpus reference was resolved through the Zenodo API, with a deliberately invalid record identifier confirmed not to resolve. The schema was checked against instances constructed to satisfy it and instances constructed to violate each conditional rule; the three limits it accepted while violated are named at 8.5. The four further limits named there follow from what JSON Schema can express and are not offered as test results.

B.6 Funding and conflicts. The work is unfunded. The author operates a consulting practice applying this framework, a conflict of interest stated here and at 11.6.

Appendix C. Figure Specifications

Figures are deposited separately and carry their own identifiers. **Each caption in the body carries the argument on its own**, so that a reader who never sees a figure loses illustration rather than content.

Figure 1, the five stages and their artifacts. The five stages in sequence, each annotated with the artifact it produces, and outbound arrows to the consumers named at 2.5. It MUST show that nothing leaves the framework that was not written by a stage, and MUST NOT show a stage out of order.

Figure 2, the residency rule set as a bounded artifact. One clause rendered with its seven required fields, inside a rule set carrying a version, an issue date and a review date. Two states MUST be distinguished: a clause within its re-verification date, and an expired clause with the rule set marked unusable for selection rather than merely stale. A superseded clause MUST be shown retained and marked.

Figure 3, the machine-readable layer and its limits. The schema file with its two profiles, drawn so that the selection points at clause identifiers and no arrow returns clause content. Above it, the seven validation requirements of 8.4, with the five the file does not reach marked as outside it and separated into the four a validator MUST perform and the one that is a person's judgement. The obligation at 8.5 that sits outside the seven MUST be drawn as outside them rather than renumbered into the list. It MUST NOT imply that a passing validation establishes conformance.

Document History

Version	Date	Summary
1.0	14 September 2026	First public version. Five ordered stages, the residency rule set as a versioned and expiring artifact, the shared machine-readable schema and its stated limits, self-declared conformance, and the naming crosswalk to the seven-layer source treatment

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About the Author

Nabeel Khan is an enterprise AI architect and governance advisor, and Founder & Solutions Architect at iSystematic Inc. He has spent twenty-five years architecting enterprise data and machine-learning systems in government, healthcare, financial services and telecommunications across three continents, grounded in TOGAF, DMBOK, ISO 27001 and SOC 2.

He works on governed production AI; systems whose authority is bounded by the architecture, whose decisions carry the evidence needed to defend them, and whose behaviour the institution can explain afterwards. The frameworks specified here, and the regulatory knowledge behind them, are published as machine-accessible AI

doi.org/10.5281/zenodo.22285047

expertise and are callable over MCP without a key at mcp.nabeelkhan.com/api/mcp, so a reader may query this specification's contents directly and receive answers carrying their source.

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```
Khan, N. A. (2026). The AI Data Governance Framework: A Five-Stage Control
  System for the Data Boundary in AI Systems.
(Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.22285047
```

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